

Akshat Sharma

BS-MS Candidate, Department of Chemical Sciences
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Research Profile

I am a chemist working on polyoxometalate and soft-oxometalate self-assembly, alongside two independent theoretical contributions: an electron-density-topology-based molecular complexity descriptor (the Quantum Assembly Index), and alongside ongoing independent work on generalized statistical complexity and variational principles for organization. My long-term goal is to develop universal theoretical frameworks for organization, self-assembly, and information processing in physical and living matter.

Research Interests

Polyoxometalate, softoxometalates, Statistical mechanics, Complex systems, Molecular self-assembly, Information theory, Origin of life, Artificial life, Active Matter, Chemical Physics, Quantum chemistry

Research Experience

Undergraduate Researcher

Dec 2023-Present

Department of Chemical Sciences, IISER Kolkata

Supervisor: **Prof. Soumyajit Roy**

- Investigated polyoxometalates, soft oxometalates, and crystallization phenomena.
- Synthesized Mo_{132} Keplerates and Mo_{154} rings; characterized using PXRD, UV-Vis, Raman.
- Studied polymorphism in pharmaceutical systems using PXRD and thermal phase analysis.
- Performed DFT and electron-density analysis for molecular structure and bonding.

Research Intern

May 2025-Jul 2025

Department of Chemistry, IIT Ropar

Supervisor: **Prof. C. M. Nagaraja**

- Synthesized NHC ligands and porous organic polymer precursors for carbon capture.
- Conducted air-sensitive synthesis and reaction monitoring under inert conditions.
- Characterized intermediates using NMR spectroscopy.

Independent Research

Electronic Structure and Bonding of Zintl Clusters

Jan 2026

Independent DFT study of electronic structure in Zintl clusters.

- Geometry optimization and electronic structure calculations (ORCA 6.0).
- Analysis of bonding in homoatomic and heteroatomic clusters.
- Shared independently with Prof. Thomas Fässler (TUM), whose group works on Zintl cluster chemistry; received a positive response.

Quantum Assembly Index

Mar 2026-May 2026

Information-theoretic framework for molecular complexity using electron density.

- Introduced Quantum Assembly Index (A_ρ).
- Benchmarked against traditional molecular complexity measures.
- ChemRxiv preprint.

Universal Core-Halo Organization of Statistical Complexity

Jun 2026-Present

Variational framework for statistical complexity in probability distributions.

- Derived universal core-halo optimal distribution.
- Demonstrated substrate-independent structure of complexity.
- Extended toward information geometry applications.

Publications & Preprints

Akshat Sharma. Quantum Assembly Theory Defines Molecular Complexity from Electron Density Topology. ChemRxiv (2026). DOI: 10.26434/chemrxiv.15000864/v3

Akshat Sharma. A Unified Discrete and Continuous Theory of Core–Halo Complexity Maximizers. Manuscript in preparation

Technical Skills

Experimental: POM synthesis ($\{\text{Mo}_{132}\}$, $\{\text{Mo}_{154}\}$), Photocatalysis, NHC ligands, POP precursors; UV-Vis, FT-IR, Raman, PXRD, SCXRD, NMR.

Computational: DFT (ORCA), Multiwfn, Avogadro, geometry optimization, frequency analysis, electron-density topology.

Software: Olex2, CrysAlisPro, Mercury, Origin, Excel, $\text{\LaTeX}$.

Programming: Python, C, Linux.

Mathematics: Linear Algebra, Real Analysis, Probability, Statistics, Group Theory, Graph Theory, Information theory.

Awards & Recognition

Reliance Foundation Undergraduate Scholarship 2024
Selected among the top 5,000 undergraduate students in India.

Leadership & Service

Office Bearer, Tennis Club 2024-2025
Indian Institute of Science Education and Research Kolkata.

Education

Indian Institute of Science Education and Research Kolkata 2023-Present
BS-MS Dual Degree Programme with major in Chemical Sciences and minor in Computational and Data Sciences.
Expected Graduation: 2028